

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1090

Cortez 138 kV
March 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1090. The Interconnection Customer proposes a 160.5 MW solar photovoltaic facility interconnecting at the Cortez 138 kV substation in the Kentucky area. The request was received on October 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 377 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,832,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Damon Substation / East Broad 138kV	Spatsum Lornex / Latah Junction 138kV line	129.0	217	8.0
Holden / Embudo 138kV	Lee County Solid Waste Energy / Cooper 345kV	115.2	236	13.6
Prairie Horizon Agri Energy / Usaa 1 138kV	Lee County Solid Waste Energy / Cooper 345kV	111.5	227	6.4
Usaa 1 Substation / Greenville Ave 138kV	Lee County Solid Waste Energy / Cooper 345kV	108.8	223	6.4
Tule / Santa Rosa Beach 138kV Line	Davisville / North Appleton 138kV	105.9	239	6.7
Rancho Carmel / Russellville East 138kV	Platte City / Albion Rd 138kV	105.3	342	7.3
W Mount Vernon / Santa Rosa Beach 138kV Line	Davisville / North Appleton 138kV line	105.3	254	6.7
Nsp Substation / West Mount Vernon 138kV	Davisville / North Appleton 138kV	104.8	270	6.7
Nsp / Wssota 138kV	North Appleton Substation / Davisville 138kV	104.2	285	6.7
Berry Street 345/138kV Auto	Norwich / Omni-Hackberry 500kV	103.0	547	13.0
Silver Bay Power / Valliant 138kV	Damon / E Broad 138kV line	101.5	203	5.4
Union / Garden Villas 138kV	Union / Desoto Pleasant Run 138kV	101.4	194	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Broad / Damon 138kV L	Rebuild line with higher-capacity conductor	\$2,975,000	11% (\$321,000)
Embudo / Holden Substation 138kV	Rebuild line with higher-capacity conductor	\$6,110,000	11% (\$698,000)
Prairie Horizon Agri Energy / Usaa 1 138kV	Rebuild line with higher-capacity conductor	\$5,069,000	13% (\$651,000)
Usaa 1 Substation / Greenville Ave 138kV	Reconductor line	\$1,639,000	13% (\$210,000)
Tule / Santa Rosa Beach 138kV Line	Reconductor line	\$2,311,000	7% (\$162,000)
Russellville East / Rancho Carmel 138kV	Reconductor line	\$3,663,000	6% (\$228,000)
W Mount Vernon / Santa Rosa Beach 138kV Line	Reconductor line	\$1,056,000	7% (\$74,000)
Nsp Substation / West Mount Vernon Substation 138kV Line	Reconductor line	\$4,791,000	7% (\$336,000)
Wissota / Nsp Substation 138kV	Reconductor line	\$4,361,000	7% (\$306,000)
Berry Street 345/138kV Auto	Replace transformer with higher-rated unit	\$11,856,000	11% (\$1,311,000)
Valliant / Silver Bay Power Substation 138kV	Reconductor line	\$2,582,000	9% (\$234,000)
Union / Garden Villas 138kV	Reconductor line	\$3,828,000	8% (\$301,000)
	Total allocated to project		\$4,832,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.